

Research Notes on Image captioning

Image captioning

>> Section 1

Automatic report generation From a commercial perspective, the most successful NLG applications have been data-to-text systems which generate textual summaries of databases and data sets; these systems usually perform data analysis as well as text generation. Research has shown that textual summaries can be more effective than graphs and other visuals for decision support, and that computer-generated texts can be superior (from the reader's perspective) to human-written texts. The first commercial data-to-text systems produced weather forecasts from weather data. The earliest such system to be deployed was FoG, which was used by Environment Canada to generate weather forecasts in French and English in the early 1990s. The success of FoG triggered other work, both research and commercial. Recent applications include the UK Met Office's text-enhanced forecast. Data-to-text systems have since been applied in a range of settings. Following the minor earthquake near Beverly Hills, California on March 17, 2014, The Los Angeles Times reported details about the time, location and strength of the quake within 3 minutes of the event. This report was automatically generated by a 'robo-journalist', which converted the incoming data into text via a preset template. Currently there is considerable commercial interest in using NLG to summarise financial and business data. Indeed, Gartner has said that NLG will become a standard feature of 90% of modern BI and analytics platforms. NLG is also being used commercially in automated journalism, chatbots, generating product descriptions for e-commerce sites, summarising medical records, and enhancing accessibility (for example by describing graphs and data sets to blind people). An example of an interactive use of NLG is the WYSIWYM framework. It stands for What you see is what you meant and allows users to see and manipulate the continuously rendered view (NLG output) of an underlying formal language document (NLG input), thereby editing the formal language without learning it. Looking ahead, the current progress in data-to-text generation paves the way for tailoring texts to specific audiences. For example, data from babies in neonatal care can be converted into text differently in a clinical setting, with different levels of technical detail and explanatory language, depending on intended recipient of the text (doctor, nurse, patient). The same idea can be applied in a sports setting, with different reports generated for fans of specific teams.

>> Section 2

Chatbots Another area where NLG has been widely applied is automated dialogue systems, frequently in the form of chatbots. A chatbot or chatterbot is a software

application used to conduct an on-line chat conversation via text or text-to-speech, in lieu of providing direct contact with a live human agent. While natural language processing (NLP) techniques are applied in deciphering human input, NLG informs the output part of the chatbot algorithms in facilitating real-time dialogues. Early chatbot systems, including Cleverbot created by Rollo Carpenter in 1988 and published in 1997, reply to questions by identifying how a human has responded to the same question in a conversation database using information retrieval (IR) techniques. Modern chatbot systems predominantly rely on machine learning (ML) models, such as sequence-to-sequence learning and reinforcement learning to generate natural language output. Hybrid models have also been explored. For example, the Alibaba shopping assistant first uses an IR approach to retrieve the best candidates from the knowledge base, then uses the ML-driven seq2seq model re-rank the candidate responses and generate the answer.

>> Section 3

Grass pollen levels for Friday have increased from the moderate to high levels of yesterday with values of around 6 to 7 across most parts of the country. However, in Northern areas, pollen levels will be moderate with values of 4.

>> Section 4

Pollen counts are expected to remain high at level 6 over most of Scotland, and even level 7 in the south east. The only relief is in the Northern Isles and far northeast of mainland Scotland with medium levels of pollen count.

>> Section 5

Evaluation As in other scientific fields, NLG researchers need to test how well their systems, modules, and algorithms work. This is called evaluation. There are three basic techniques for evaluating NLG systems:

>> Section 6

Task-based (extrinsic) evaluation: give the generated text to a person, and assess how well it helps them perform a task (or otherwise achieves its communicative goal). For example, a system which generates summaries of medical data can be evaluated by giving these summaries to doctors, and assessing whether the summaries help doctors make better decisions. **Human ratings:** give the generated text to a person, and ask them to rate the quality and usefulness of the text. **Metrics:** compare generated texts to texts written by people from the same input data, using an automatic metric such as BLEU, METEOR, ROUGE and LEPOR. An ultimate goal is how useful NLG systems are at helping people, which is the first of the

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above techniques. However, task-based evaluations are time-consuming and expensive, and can be difficult to carry out (especially if they require subjects with specialised expertise, such as doctors). Hence (as in other areas of NLP) task-based evaluations are the exception, not the norm. Recently researchers are assessing how well human-ratings and metrics correlate with (predict) task-based evaluations. Work is being conducted in the context of Generation Challenges shared-task events. Initial results suggest that human ratings are much better than metrics in this regard. In other words, human ratings usually do predict task-effectiveness at least to some degree (although there are exceptions), while ratings produced by metrics often do not predict task-effectiveness well. These results are preliminary. In any case, human ratings are the most popular evaluation technique in NLG; this is contrast to machine translation, where metrics are widely used. An AI can be graded on faithfulness to its training data or, alternatively, on factuality. A response that reflects the training data but not reality is faithful but not factual. A confident but unfaithful response is a hallucination. In Natural Language Processing, a hallucination is often defined as "generated content that is nonsensical or unfaithful to the provided source content".

>> Section 7

Natural language generation (NLG) is a software process that produces natural language output. A widely cited survey of NLG methods describes NLG as "the subfield of artificial intelligence and computational linguistics that is concerned with the construction of computer systems that can produce understandable texts in English or other human languages from some underlying non-linguistic representation of information". While it is widely agreed that the output of any NLG process is text, there is some disagreement about whether the inputs of an NLG system need to be non-linguistic. Common applications of NLG methods include the production of various reports, for example weather and patient reports; image captions; and chatbots like ChatGPT. Automated NLG can be compared to the process humans use when they turn ideas into writing or speech. Psycholinguists prefer the term language production for this process, which can also be described in mathematical terms, or modeled in a computer for psychological research. NLG systems can also be compared to translators of artificial computer languages, such as decompilers or transpilers, which also produce human-readable code generated from an intermediate representation. Human languages tend to be considerably more complex and allow for much more ambiguity and variety of expression than programming languages, which makes NLG more challenging. NLG may be viewed as complementary to natural language

understanding (NLU): whereas in natural-language understanding, the system needs to disambiguate the input sentence to produce the machine representation language, in NLG the system needs to make decisions about how to put a representation into words. The practical considerations in building NLU vs. NLG systems are not symmetrical. NLU needs to deal with ambiguous or erroneous user input, whereas the ideas the system wants to express through NLG are generally known precisely. NLG needs to choose a specific, self-consistent textual representation from many potential representations, whereas NLU generally tries to produce a single, normalized representation of the idea expressed. NLG has existed since ELIZA was developed in the mid 1960s, but the methods were first used commercially in the 1990s. NLG techniques range from simple template-based systems like a mail merge that generates form letters, to systems that have a complex understanding of human grammar. NLG can also be accomplished by training a statistical model using machine learning, typically on a large corpus of human-written texts.